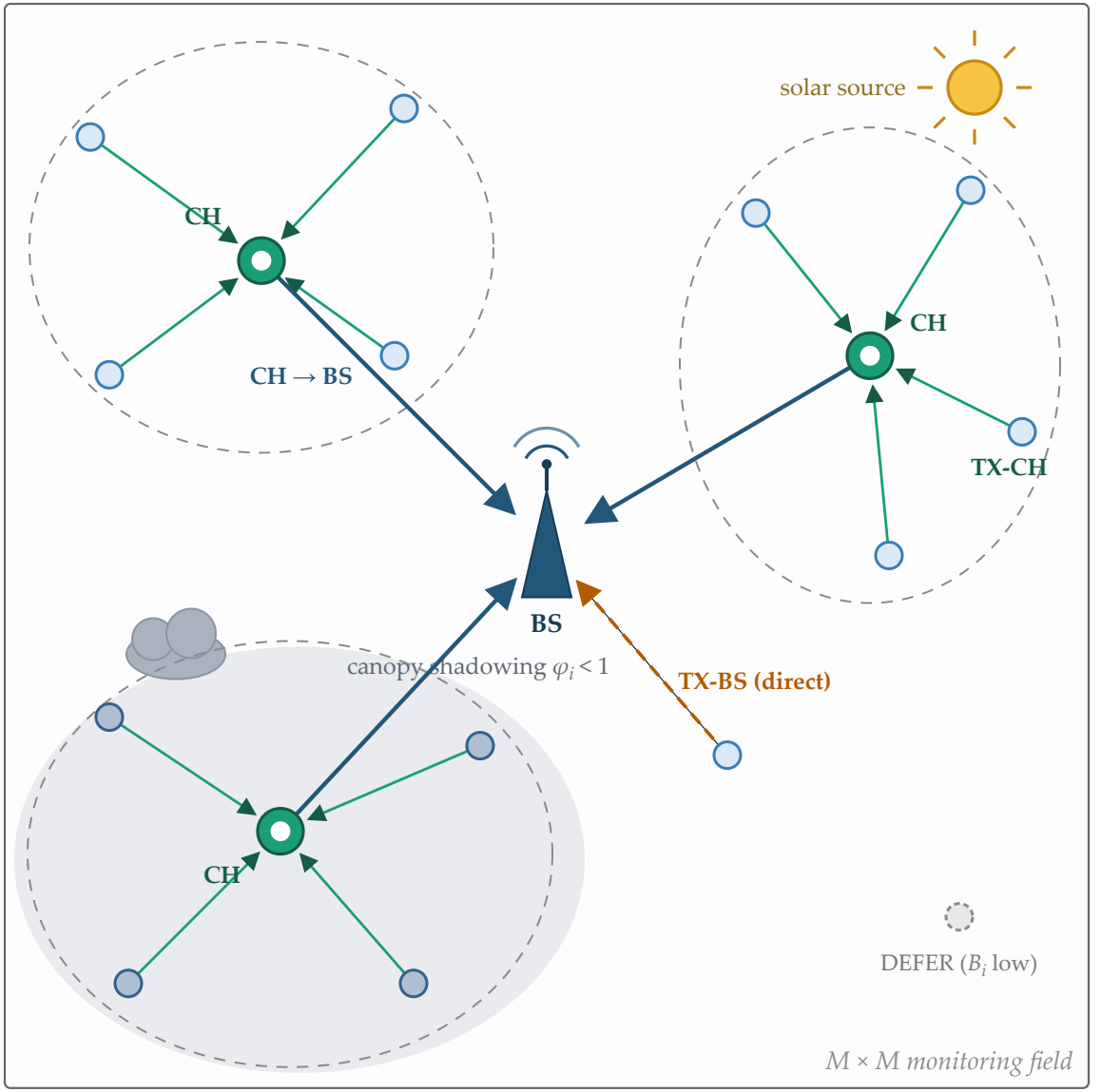


Legend

- Base station (BS)
- Cluster head (CH)
- Member sensor node
- Shadowed node (low φ_i)
- Depleted node (defer)
- TX-CH: member → CH
- CH forwarding to BS
- TX-BS: direct transmission
- Cluster boundary
- Shadowed region
- Solar harvesting source